

WI4014TU – Numerical Analysis for PDE's

Lecture 14

Linear solvers, LU factorization, classical iterative methods, regular splittings, Jacobi, Gauss-Seidel, SOR, Gerschgorin's Lemma

Why study linear solvers?

- Numerical solution of all PDE's involves solving one or multiple linear algebraic problems: Poisson eq. (one), diffusion with Backward Euler (one per outer iteration), nonlinear equations with BE (one per inner iteration), etc
- Matrix assembly and linear solvers are computationally expensive
- Large computational domains and fine spatial discretization lead to very large linear algebraic problems
- Some system matrices are symmetric, other are not and may require special linear solvers

Well-conditioned linear problems

We are considering linear systems of the form

$$A\mathbf{x} = \mathbf{b}$$

where

- $A \in \mathbb{R}^{n \times n}$, $\mathbf{x}, \mathbf{b} \in \mathbb{R}^n$
- A is *nonsingular* and *well conditioned*
- we shall see how other properties of A help to find a suitable solver

A few words about direct solvers

LU factorization

- A well conditioned system $A\mathbf{x} = \mathbf{b}$ can, in principle, be solved by Gaussian elimination without pivoting
- A practical way of doing Gaussian elimination is to factorize the matrix A as:

$$A = LU$$

where U is upper triangular and L is lower-triangular with all diagonal elements equal unity

- Then, the linear system $A\mathbf{x} = \mathbf{b}$ is replaced by two systems $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$ both of which can be solved by simple substitution in $\mathcal{O}(n^2)$ Floating-Point Operations (FLOP's)
- LU factorization itself costs $\mathcal{O}(n^3)$ operations:

$$u_{k,j} = a_{k,j} - \sum_{i=1}^{j-1} \ell_{k,i} u_{i,j}, \quad j = k, k+1, \dots, n$$

$$\ell_{j,k} = \frac{1}{u_{k,k}} \left(a_{j,k} - \sum_{i=1}^{k-1} \ell_{j,i} u_{i,k} \right), \quad j = k+1, k+2, \dots, n$$

LU factorization of banded matrices

- Matrix A with elements $a_{k,m}$ is called *banded* with *bandwidth* s if $a_{k,m} = 0$ for every $k, m \in \{1, 2, \dots, n\}$ such that $|k - m| > s$.
- Example: tri-diagonal matrix has bandwidth $s = 1$
- Ordering of unknowns influences the bandwidth of A
- **Storage (elements):** dense – n^2 , banded $\sim (2s + 1)n$
- **Operations count (flops):**
 - ▶ LU factorization: dense – $\mathcal{O}(n^3)$, banded – $\mathcal{O}(s^2 n)$
 - ▶ Solving $Ly = b$ and $Ux = y$: dense – $\mathcal{O}(n^2)$, banded – $\mathcal{O}(sn)$
- If the matrix A is positive definite, then a twice cheaper ‘perfect’ Cholesky factorization $A = LL^T$ can be applied if the ordering of unknowns is right (tree structure)
- Solver based on factorization of A is called a *direct solver*

Classical Iterative Methods

Linear one-step stationary iterative scheme

Consider the system

$$A\mathbf{x} = \mathbf{b} \quad (1)$$

and the iterative procedure:

$$\mathbf{x}_{k+1} = H\mathbf{x}_k + \mathbf{v} \quad (2)$$

The *spectral radius* $\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}$, where $\sigma(A)$ is the set of eigenvalues (*spectrum*) of A .

It is always true that $\rho(A) \leq \|A\|$, where $\|\cdot\|$ is the Euclidean norm. For *normal* matrices, i.e., such that $A^T A = A A^T$, $\rho(A) = \|A\|$. For general square matrices: $\|A\| = [\rho(A^T A)]^{1/2}$

Theorem: Given an arbitrary linear system (1), a linear, one-step, stationary scheme (2) converges to a unique bounded limit $\tilde{\mathbf{x}}$, regardless of the choice of the starting vector $\mathbf{x}_0$, if and only if $\rho(H) < 1$. Moreover, $\tilde{\mathbf{x}}$ is the solution of (1) if and only if $\mathbf{v} = (I - H)A^{-1}\mathbf{b}$

Proof (1): Let $\rho(H) < 1$ and let H be diagonalizable with the eigendecomposition $H = V\Lambda V^{-1}$. Then,

$$\begin{aligned}\lim_{k \rightarrow \infty} H^k &= \lim_{k \rightarrow \infty} (V\Lambda V^{-1})^k = \lim_{k \rightarrow \infty} (V\Lambda V^{-1} V\Lambda V^{-1} \dots V\Lambda V^{-1}) \\ &= V \left(\lim_{k \rightarrow \infty} \Lambda^k \right) V^{-1} = 0\end{aligned}$$

since $|\lambda_i| < 1$, $\lambda_i \in \sigma(H)$. If H is not diagonalizable, the same can be proven using the Jordan decomposition $H = WJW^{-1}$.

We claim that, in general:

$$\mathbf{x}_k = H^k \mathbf{x}_0 + (I - H)^{-1}(I - H^k) \mathbf{v}, \quad k = 0, 1, \dots$$

Induction: (i) It is true for $k = 0$, since $H^0 = I$. (ii) Assume it holds for some $k \geq 0$. (iii) Then, it also holds for $k + 1$:

$$\begin{aligned}\mathbf{x}_{k+1} &= H\mathbf{x}_k + \mathbf{v} = H \left[H^k \mathbf{x}_0 + (I - H)^{-1}(I - H^k) \mathbf{v} \right] + \mathbf{v} \\ &= H^{k+1} \mathbf{x}_0 + \left[H(I - H)^{-1}(I - H^k) + I \right] \mathbf{v} \\ &= H^{k+1} \mathbf{x}_0 + \left[(I - H)^{-1}H(I - H^k) + (I - H)^{-1}(I - H) \right] \mathbf{v} \\ &= H^{k+1} \mathbf{x}_0 + (I - H)^{-1}(H - H^{k+1} + I - H) \mathbf{v} \\ &= H^{k+1} \mathbf{x}_0 + (I - H)^{-1}(I - H^{k+1}) \mathbf{v}\end{aligned}$$

Proof (2): Hence, for $\rho(H) < 1$, the iterative process converges to the vector $\tilde{\mathbf{x}}$:

$$\tilde{\mathbf{x}} = \lim_{k \rightarrow \infty} \mathbf{x}_k = \lim_{k \rightarrow \infty} H^k \mathbf{x}_0 + (I - H)^{-1}(I - H^k) \mathbf{v} = (I - H)^{-1} \mathbf{v}$$

Let $\rho(H) \geq 1$.

If $1 \notin \sigma(H)$, then $I - H$ is invertible and $\tilde{\mathbf{x}} = (I - H)^{-1} \mathbf{v}$ is the only possible bounded limit of the iterative scheme. Assume there was another bounded limit $\tilde{\mathbf{y}}$. Then,

$$\tilde{\mathbf{y}} = \lim_{k \rightarrow \infty} \mathbf{x}_{k+1} = \lim_{k \rightarrow \infty} (H \mathbf{x}_k + \mathbf{v}) = H \lim_{k \rightarrow \infty} \mathbf{x}_k + \mathbf{v} = H \tilde{\mathbf{y}} + \mathbf{v}$$

and $\tilde{\mathbf{y}} = (I - H)^{-1} \mathbf{v} = \tilde{\mathbf{x}}$, i.e., the limit, if it exists, is unique.

If $1 \in \sigma(H)$ and $\mathbf{w}$ is the corresponding eigenvector $H \mathbf{w} = \mathbf{w}$, then, whenever there is a limit, $\tilde{\mathbf{y}} = H \tilde{\mathbf{y}} + \mathbf{v}$, it follows that $\tilde{\mathbf{y}} + \mathbf{w} = H(\tilde{\mathbf{y}} + \mathbf{w}) + \mathbf{v}$ is also a limit. Hence, with $1 \in \sigma(H)$, either there is no limit or the limit is not unique, i.e., there is no convergence.

Proof (3):

Choose $\lambda \in \sigma(H)$ such that $|\lambda| = \rho(H) \geq 1$. Let now $\mathbf{w}$ be the eigenvector of this eigenvalue, i.e., $H\mathbf{w} = \lambda\mathbf{w}$, and let $\|\mathbf{w}\| = 1$. We show that in this case there is always a starting vector $\mathbf{x}_0$ for which the iterations will fail to converge.

Case 1: Let $\lambda \in \mathbb{R}$. Hence, $\mathbf{w}$ is also real (up to a multiplier). Choose the starting vector as $\mathbf{x}_0 = \mathbf{w} + \tilde{\mathbf{x}}$. We claim that

$$\mathbf{x}_k = \lambda^k \mathbf{w} + \tilde{\mathbf{x}}, \quad k = 0, 1, \dots$$

Induction: (i) This is true for $k = 0$. (ii) Assume it is true for some $k \geq 0$. (iii) Then, it is also true for $k + 1$:

$$\begin{aligned} \mathbf{x}_{k+1} &= H\mathbf{x}_k + \mathbf{v} = H(\lambda^k \mathbf{w} + \tilde{\mathbf{x}}) + \mathbf{v} = \lambda^{k+1} \mathbf{w} + H(I - H)^{-1} \mathbf{v} + \mathbf{v} \\ &= \lambda^{k+1} \mathbf{w} + (I - H)^{-1} H \mathbf{v} + (I - H)^{-1} (I - H) \mathbf{v} \\ &= \lambda^{k+1} \mathbf{w} + (I - H)^{-1} (H + I - H) \mathbf{v} = \lambda^{k+1} \mathbf{w} + (I - H)^{-1} \mathbf{v} = \lambda^{k+1} \mathbf{w} + \tilde{\mathbf{x}} \end{aligned}$$

It follows that, since $|\lambda| \geq 1$,

$$\|\mathbf{x}_k - \tilde{\mathbf{x}}\| = \|\lambda^k \mathbf{w}\| = |\lambda|^k \geq 1, \quad k = 0, 1, \dots$$

In other words, the sequence $\{\mathbf{x}_k\}$ does not converge to $\tilde{\mathbf{x}}$.

Proof (4): Case 2: Let $\lambda \in \mathbb{C}$. Then, if $(\lambda, \mathbf{w})$ is an eigenpair, so is $(\bar{\lambda}, \bar{\mathbf{w}})$. Furthermore, since $\lambda \neq \bar{\lambda}$, the two eigenvectors, $\mathbf{w}$ and $\bar{\mathbf{w}}$, are linearly independent. Otherwise, they would correspond to the same eigenvalue. Consider the function

$$g(z) = \|z\mathbf{w} + \bar{z}\bar{\mathbf{w}}\|, \quad z \in \mathbb{C}$$

This is a continuous function $g : \mathbb{C} \rightarrow \mathbb{R}$. Hence, it attains a minimum on every closed and bounded subset of $\mathbb{C}$, in particular, on the unit circle. Therefore,

$$\inf_{-\pi \leq \theta \leq \pi} \|e^{i\theta}\mathbf{w} + e^{-i\theta}\bar{\mathbf{w}}\| = \min_{-\pi \leq \theta \leq \pi} \|e^{i\theta}\mathbf{w} + e^{-i\theta}\bar{\mathbf{w}}\| = \nu \geq 0$$

Suppose that $\nu = 0$. Then there exists $\theta_0 \in [-\pi, \pi]$ such that

$$\|e^{i\theta_0}\mathbf{w} + e^{-i\theta_0}\bar{\mathbf{w}}\| = 0$$

and $\bar{\mathbf{w}} = e^{2i\theta_0}\mathbf{w}$, i.e., $\mathbf{w}$ and $\bar{\mathbf{w}}$ are linearly dependent, which is not possible. Hence, $\nu > 0$. The function $g(z)$ is homogeneous,

$$g(re^{i\theta}) = \|re^{i\theta}\mathbf{w} + re^{-i\theta}\bar{\mathbf{w}}\| = r\|e^{i\theta}\mathbf{w} + e^{-i\theta}\bar{\mathbf{w}}\| = rg(e^{i\theta}), \quad r > 0, \quad |\theta| \leq \pi$$

Proof (5): In particular,

$$g(z) = g\left(|z| \frac{z}{|z|}\right) = |z| g\left(\frac{z}{|z|}\right) \geq |z| \nu, \quad z \in \mathbb{C} \setminus \{0\}$$

Choose the starting vector as $\mathbf{x}_0 = \mathbf{w} + \overline{\mathbf{w}} + \tilde{\mathbf{x}}$. It is possible to show inductively that

$$\mathbf{x}_k = \lambda^k \mathbf{w} + \overline{\lambda}^k \overline{\mathbf{w}} + \tilde{\mathbf{x}}, \quad k = 0, 1, \dots$$

Hence,

$$\|\mathbf{x}_k - \tilde{\mathbf{x}}\| = \|\lambda^k \mathbf{w} + \overline{\lambda}^k \overline{\mathbf{w}}\| = g(\lambda^k) \geq |\lambda|^k \nu \geq \nu > 0, \quad k = 0, 1, \dots$$

and the sequence $\{\mathbf{x}_k\}$ is bounded away from the only possible limit $\tilde{\mathbf{x}}$, i.e., it cannot converge.

The claim that $\mathbf{v}$ must be equal $(I - H)A^{-1}\mathbf{b}$ is easy to check, since $\mathbf{x} = A^{-1}\mathbf{b}$ is the exact solution and $\tilde{\mathbf{x}} = (I - H)^{-1}\mathbf{v}$ is the unique limit. $\square$

One-step iteration: conclusion

- To solve $A\mathbf{x} = \mathbf{b}$
- One could use iterations $\mathbf{x}_{k+1} = H\mathbf{x}_k + \mathbf{v}$
- Provided that $\rho(H) < 1$ and $\mathbf{v} = (I - H)A^{-1}\mathbf{b}$
- Not practical, used as a tool for studying other methods

Incomplete LU factorization (ILU)

Suppose that there is a splitting $A = \hat{A} - E$ such that it is easy to find the LU factorization of $\hat{A}$ (e.g. $\hat{A}$ is banded). Then,

$$Ax = b$$

$$(\hat{A} - E)x = b$$

$$\hat{A}x = Ex + b$$

This suggest the following *Incomplete LU Factorization* iterative scheme:

$$\hat{A}x_{k+1} = Ex_k + b$$

where one needs to compute the LU factorization of $\hat{A}$ and re-use it at each iteration to find x_{k+1} .

This can be put in the form $x_{k+1} = Hx_k + v$ if we choose $H = \hat{A}^{-1}E$ and $v = \hat{A}^{-1}b$

Regular splittings

ILU, where $A = \hat{A} - E$, is an example of the *regular splitting* $A = P - N$, where P is *nonsingular*. Consider the iterative scheme:

$$P\mathbf{x}_{k+1} = N\mathbf{x}_k + \mathbf{b} \quad (3)$$

Theorem: Suppose that both A and $P + P^T - A$ are real, symmetric and positive definite (SPD). Then the scheme (3) converges.

Note that in this case $H = P^{-1}N = P^{-1}(P - A) = I - P^{-1}A$ and $\mathbf{v} = P^{-1}\mathbf{b}$. Proof shows that under these assumptions all eigenvalues λ of $I - P^{-1}A$ have $|\lambda| < 1$.

Proof (1): Let $\lambda \in \mathbb{C}$ be an arbitrary eigenvalue of $H = I - P^{-1}A$ and $\mathbf{w}$ – the corresponding eigenvector. Then,

$$(I - P^{-1}A)\mathbf{w} = \lambda\mathbf{w}$$

$$P(I - P^{-1}A)\mathbf{w} = P\lambda\mathbf{w}$$

$$(P - A)\mathbf{w} = \lambda P\mathbf{w}$$

$$P\mathbf{w} - \lambda P\mathbf{w} = A\mathbf{w}$$

$$(1 - \lambda)P\mathbf{w} = A\mathbf{w}$$

First, we conclude that $\lambda \neq 1$, since otherwise $A\mathbf{w} = \mathbf{0}$ for some $\mathbf{w} \neq \mathbf{0}$, i.e., A is singular, which contradicts the assumption that A is SPD. Further, since A is symmetric and $\lambda \neq 1$,

$$(1 - \lambda)\overline{\mathbf{w}}^T P\mathbf{w} = \overline{\mathbf{w}}^T A\mathbf{w} \in \mathbb{R}$$

$$(1 - \overline{\lambda})\overline{\mathbf{w}}^T P^T \mathbf{w} = \overline{\mathbf{w}}^T A\mathbf{w}$$

$$\left(\frac{1}{1 - \lambda} + \frac{1}{1 - \overline{\lambda}} - 1 \right) \overline{\mathbf{w}}^T A\mathbf{w} = \overline{\mathbf{w}}^T (P + P^T - A)\mathbf{w}$$

Proof (2): Since

$$\frac{1}{1-\lambda} + \frac{1}{1-\bar{\lambda}} - 1 = \frac{1-|\lambda|^2}{|1-\lambda|^2} \in \mathbb{R}$$

is real, but $\mathbf{w}$ may be complex $\mathbf{w} = \mathbf{w}_R + i\mathbf{w}_I$, we take the real part

$$\operatorname{Re} \left\{ \left(\frac{1}{1-\lambda} + \frac{1}{1-\bar{\lambda}} - 1 \right) \bar{\mathbf{w}}^T A \mathbf{w} \right\} = \operatorname{Re} \{ \bar{\mathbf{w}}^T (P + P^T - A) \mathbf{w} \}$$

We compute

$$\operatorname{Re} \{ \bar{\mathbf{w}}^T A \mathbf{w} \} = \operatorname{Re} \{ (\mathbf{w}_R - i\mathbf{w}_I)^T A (\mathbf{w}_R + i\mathbf{w}_I) \} = \mathbf{w}_R^T A \mathbf{w}_R + \mathbf{w}_I^T A \mathbf{w}_I$$

$$\operatorname{Re} \{ \bar{\mathbf{w}}^T (P + P^T - A) \mathbf{w} \} = \mathbf{w}_R^T (P + P^T - A) \mathbf{w}_R + \mathbf{w}_I^T (P + P^T - A) \mathbf{w}_I$$

So that

$$\frac{1-|\lambda|^2}{|1-\lambda|^2} (\mathbf{w}_R^T A \mathbf{w}_R + \mathbf{w}_I^T A \mathbf{w}_I) = \mathbf{w}_R^T (P + P^T - A) \mathbf{w}_R + \mathbf{w}_I^T (P + P^T - A) \mathbf{w}_I$$

Proof (3): Both A and $P + P^T - A$ are positive definite and it is impossible for both $\mathbf{w}_R$ and $\mathbf{w}_I$ to vanish simultaneously, since this would imply $\mathbf{w} = \mathbf{0}$. Hence,

$$\mathbf{w}_R^T A \mathbf{w}_R + \mathbf{w}_I^T A \mathbf{w}_I > 0$$

$$\mathbf{w}_R^T (P + P^T - A) \mathbf{w}_R + \mathbf{w}_I^T (P + P^T - A) \mathbf{w}_I > 0$$

Therefore,

$$\frac{1 - |\lambda|^2}{|1 - \lambda|^2} > 0$$

Hence, $|\lambda| < 1$ for any $\lambda \in \sigma(H)$. Consequently, $\rho(H) < 1$ and the iterative scheme (3) converges. $\square$

Jacobi, Gauss-Seidel, and SOR methods

Any matrix A can be split as $A = D - L_0 - U_0$, where D is diagonal, L_0 is strictly lower-triangular, and U_0 is strictly upper-triangular. Let $L = D^{-1}L_0$ and $U = D^{-1}U_0$. Then, an iterative scheme $\mathbf{x}_{k+1} = H\mathbf{x}_k + \mathbf{v}$ can be constructed in several new ways:

- *Jacobi* method: $H = L + U$ and $\mathbf{v} = D^{-1}\mathbf{b}$ or as regular splitting

$$D\mathbf{x}_{k+1} = (L_0 + U_0)\mathbf{x}_k + \mathbf{b}$$

- *Gauss-Seidel* method: $H = (I - L)^{-1}U$ and $\mathbf{v} = (I - L)^{-1}D^{-1}\mathbf{b}$ or as regular splitting

$$(D - L_0)\mathbf{x}_{k+1} = U_0\mathbf{x}_k + \mathbf{b}$$

- *Successive Over-Relaxation* (SOR) method:
 $H_\omega = (I - \omega L)^{-1}[(1 - \omega)I + \omega U]$ and $\mathbf{v}_\omega = \omega(I - \omega L)^{-1}D^{-1}\mathbf{b}$, where $\omega \in [1, 2)$ is the *relaxation parameter*. As regular splitting:

$$\left(\frac{1}{\omega}D - L_0\right)\mathbf{x}_{k+1} = \left[\left(\frac{1}{\omega} - 1\right)D + U_0\right]\mathbf{x}_k + \mathbf{b}$$

A Tool to Estimate the Location of Eigenvalues

Lemma (Gerschgorin Circles)

Let $A \in \mathbb{C}^{n \times n}$ be an arbitrary matrix with elements $a_{k,m}$. The spectrum $\sigma(A)$ is contained within the union of n disks:

$$\sigma(A) \subset \bigcup_{k=1}^n \mathbb{D}_k \quad (4)$$

where each Gerschgorin disk $\mathbb{D}_k$ is defined in the complex plane as:

$$\mathbb{D}_k = \left\{ z \in \mathbb{C} : |z - a_{k,k}| \leq \sum_{m \neq k}^n |a_{k,m}| \right\} \quad (5)$$

Boundary Property for Irreducible Matrices

If A is **irreducible** (e.g., connected grid graphs in FDM/FVM) and an eigenvalue $\lambda \in \sigma(A)$ lies on the boundary of the total union $\partial(\bigcup \mathbb{D}_k)$, then λ must lie on the boundary $\partial \mathbb{D}_k$ for **all** $k = 1, \dots, n$.

Example: FDM/FVM Laplacian, uniform mesh, $k = 1$

Inner point FVM equation:

$$-\frac{1}{h_x^2}u_{i-1,j} - \frac{1}{h_y^2}u_{i,j-1} + \left(\frac{2}{h_x^2} + \frac{2}{h_y^2}\right)u_{i,j} - \frac{1}{h_x^2}u_{i+1,j} - \frac{1}{h_y^2}u_{i,j+1} = f_{ij}$$

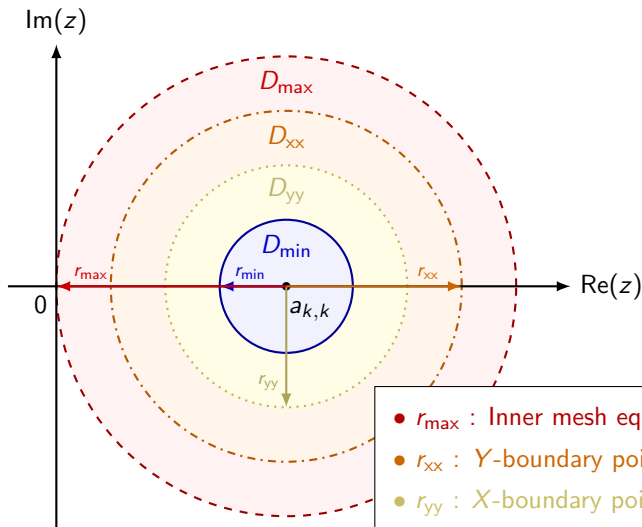
Since the main diagonal is constant, the centers of all Gerschgorin disks are the same:

$$a_{k,k} = \frac{2}{h_x^2} + \frac{2}{h_y^2}, \quad k = 1, \dots, (N_x - 1)(N_y - 1)$$

The radii of disks have four possible values:

$$\begin{aligned} r_{\min} &= \frac{1}{h_x^2} + \frac{1}{h_y^2}, & r_{\max} &= \frac{2}{h_x^2} + \frac{2}{h_y^2} \\ r_{yy} &= \frac{1}{h_x^2} + \frac{2}{h_y^2}, & r_{xx} &= \frac{2}{h_x^2} + \frac{1}{h_y^2} \end{aligned}$$

Example: $h_x < h_y$



- $r_{\max}$: Inner mesh equations (touches 0)
- r_{xx} : Y-boundary points
- r_{yy} : X-boundary points
- $r_{\min}$: Corner boundary points

Bounds on minimal and maximal eigenvalues

- Since the matrix is symmetric, all eigenvalues are real – on the $\text{Re}(z)$ axis.
- The smallest eigenvalue is at the leftmost possible location along the $\text{Re}(z)$ axis. However, it cannot be at zero, since in that case all eigenvalues must be zero as we have an irreducible matrix! Gerschgorin's Lemma shows that $\lambda_{\min} > 0$
- The largest eigenvalue is at the rightmost location along $\text{Re}(z)$. The algebraic bound:

$$\lambda_{\max} \leq a_{k,k} + r_{\max} = \frac{4}{h_x^2} + \frac{4}{h_y^2}$$

Definition: A matrix A is (strictly) *diagonally dominant* if

$$|a_{k,k}| \geq \sum_{m \neq k}^n |a_{k,m}|, \quad k = 1, 2, \dots, n \quad (6)$$

with strict inequality for at least one row.

Theorem: *If the matrix A is irreducible and diagonally dominant, then both the Jacobi and Gauss-Seidel methods converge.*

Proof (1): Jacobi: The iteration matrix of the Jacobi method is $H = D^{-1}L_0 + D^{-1}U_0$, where $d_{k,k} = a_{k,k}$. Hence,

$$\sum_{m=1}^n |h_{k,m}| = \sum_{m \neq k}^n |h_{k,m}| = \frac{1}{|a_{k,k}|} \sum_{m \neq k}^n |a_{k,m}| \leq 1, \quad k = 1, 2, \dots, n$$

and this inequality is sharp for at least one k . Since $h_{k,k} = 0$, by the Gerschgorin Lemma, $\rho(H) < 1$ and the Jacobi iteration converges.

Gauss-Seidel: For the Gauss-Seidel method: $H = (I - D^{-1}L_0)^{-1}D^{-1}U_0$, $\mathbf{v} = (I - D^{-1}L_0)^{-1}D^{-1}\mathbf{b}$. Choose $\lambda \in \sigma(H)$ with the corresponding eigenvector $\mathbf{w}$. Hence,

$$\begin{aligned} H\mathbf{w} &= \lambda\mathbf{w} \\ (I - D^{-1}L_0)^{-1}D^{-1}U_0\mathbf{w} &= \lambda\mathbf{w} \\ U_0\mathbf{w} &= \lambda D(I - D^{-1}L_0)\mathbf{w} \\ U_0\mathbf{w} &= \lambda(D - L_0)\mathbf{w} \end{aligned}$$

Proof (2): Since $A = D - L_0 - U_0$, we can rewrite $U_0 \mathbf{w} = \lambda(D - L_0) \mathbf{w}$ as

$$\begin{aligned} - \sum_{m=k+1}^n a_{k,m} w_m &= \lambda a_{k,k} w_k + \lambda \sum_{m=1}^{k-1} a_{k,m} w_m \\ \lambda a_{k,k} w_k &= - \sum_{m=k+1}^n a_{k,m} w_m - \lambda \sum_{m=1}^{k-1} a_{k,m} w_m \\ |\lambda| |a_{k,k}| |w_k| &\leq \sum_{m=k+1}^n |a_{k,m}| |w_m| + |\lambda| \sum_{m=1}^{k-1} |a_{k,m}| |w_m| \\ &\leq \left(\sum_{m=k+1}^n |a_{k,m}| + |\lambda| \sum_{m=1}^{k-1} |a_{k,m}| \right) \max_m |w_m| \end{aligned}$$

for all $k = 1, 2, \dots, n$

Proof (3): Let q be the index for which $|w_q| = \max_m |w_m| > 0$. Then,

$$|\lambda| |a_{q,q}| \leq \sum_{m=q+1}^n |a_{q,m}| + |\lambda| \sum_{m=1}^{q-1} |a_{q,m}|$$

Assume that $|\lambda| \geq 1$. In which case

$$|a_{q,q}| \leq \frac{1}{|\lambda|} \sum_{m=q+1}^n |a_{q,m}| + \sum_{m=1}^{q-1} |a_{q,m}| \leq \sum_{m=q+1}^n |a_{q,m}| + \sum_{m=1}^{q-1} |a_{q,m}| \leq \sum_{m \neq q}^n |a_{q,m}|$$

However, since also $|a_{q,q}| \geq \sum_{m \neq q}^n |a_{q,m}|$, it must be that $|a_{q,q}| = \sum_{m \neq q}^n |a_{q,m}|$.
At the same time, as we showed earlier

$$|\lambda| |a_{k,k}| |w_k| \leq \sum_{m=k+1}^n |a_{k,m}| |w_m| + |\lambda| \sum_{m=1}^{k-1} |a_{k,m}| |w_m|, \quad k = 1, 2, \dots, n$$

Proof (4): Considering $k = q$ and substituting $|a_{q,q}| = \sum_{m \neq q}^n |a_{q,m}|$ we get

$$\begin{aligned}
 |\lambda| |a_{q,q}| |w_q| &\leq \sum_{m=q+1}^n |a_{q,m}| |w_m| + |\lambda| \sum_{m=1}^{q-1} |a_{q,m}| |w_m| \\
 |\lambda| \sum_{m \neq q}^n |a_{q,m}| |w_q| &\leq \sum_{m=q+1}^n |a_{q,m}| |w_m| + |\lambda| \sum_{m=1}^{q-1} |a_{q,m}| |w_m| \leq |\lambda| \sum_{m \neq q}^n |a_{q,m}| |w_m| \\
 \sum_{m \neq q}^n |a_{q,m}| |w_q| &\leq \sum_{m \neq q}^n |a_{q,m}| |w_m|
 \end{aligned}$$

Since $|w_q| = \max_m |w_m|$, the last inequality can only hold if $|w_m| = |w_q|$ for all $m = 1, 2, \dots, n$. Hence, any $k = 1, 2, \dots, n$ can be chosen as q and the equality $|a_{k,k}| = \sum_{m \neq k}^n |a_{k,m}|$ must hold for all $k = 1, 2, \dots, n$, which contradicts the assumption of strict diagonal dominance. Therefore, $|\lambda| < 1$ and $\rho(H) < 1$. $\square$

Classical iterative methods: conclusion

- To solve $A\mathbf{x} = \mathbf{b}$
- One could use various regular splittings $A = P - N$
- And iterations $P\mathbf{x}_{k+1} = N\mathbf{x}_k + \mathbf{b}$, which is the same as $\mathbf{x}_{k+1} = P^{-1}N\mathbf{x}_k + P^{-1}\mathbf{b}$
- There are several splittings, which guarantee $\rho(P^{-1}N) < 1$, i.e. convergence
- Sometimes a symmetry of A is assumed, convergence may be slow